

NAVYA SAXENA

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EDUCATION

Pimpri Chinchwad College of Engineering (PCCOE)

B.Tech in Computer Engineering | CGPA: 8.25

Pune, Maharashtra

Graduation: May 2027

WORK EXPERIENCE

Electrofine (E-Waste Management NGO)

Feb 2025 – April 2025

- Built and shipped E-Link, a multi-role e-waste management web platform connecting users with recyclers, NGOs, repair experts, and refurbishers.
- Implemented multi-role authentication, pickup scheduling features to streamline the end-to-end e-waste disposal
- Developed educational modules, recycling-awareness content, and a refurbishment marketplace module, supporting the NGO's circular-economy mission and driving user engagement.

CESA ACM-W PCCOE

Jr. Secretary

Sep 2024 – Aug 2025

- Coordinated a series of 12 technical events end-to-end, collaborating across a diverse 30-member team to strengthen engagement, planning, and cross-functional execution.
- Spearheaded an all-women hackathon during the college techfest, driving participation from 200+ attendees and mentoring teams whose solutions were selected for funding — demonstrating leadership, communication, and real-world problem-solving under time constraints.

PROJECTS

Predictive Handover Framework (LiFi-WiFi) — Patent Application Filed

Aug 2025 – Nov 2025

- Engineered a predictive handover framework enabling seamless Li-Fi/Wi-Fi connectivity by analyzing smartphone accelerometer and gyroscope data, tackling a real-world limitation in hybrid wireless networks.
- Designed and trained LSTM and Transformer models, achieving ~87% and ~90% prediction accuracy respectively, outperforming traditional RSS-based handover methods through rigorous analytical evaluation.
- Deployed the model via a Flask API and validated results through live simulation testing; filed a patent application for the framework, reflecting innovation and technical depth.

Event Registration Platform

Jan 2026 – Feb 2026

- Built a full-stack event registration platform for large-scale hackathons and college fests, implementing role-based authentication across organizer and participant portals.
- Developed dynamic registration workflows with real-time form handling, automated participant ID generation, and structured relational-database storage for event and user data.
- Architected the platform to support thousands of concurrent users during peak registration windows using auto-scaling infrastructure, ensuring zero downtime — applying core software engineering and system design principles.

Attendance Monitoring & Automated Warning-Letter Generation

March 2026

- Developed a web-based attendance monitoring system to parse XLSX reports and automatically flag students below attendance thresholds, streamlining academic administration.
- Implemented dynamic DOCX warning-letter generation using JavaScript and JSZip, eliminating manual documentation effort and reducing turnaround time.
- Added bulk student selection and automated email notifications via Node.js, improving communication efficiency and adaptability of the tool to institutional needs.

TECHNICAL SKILLS

Languages: Java, C, C++, Python, R

Frameworks: Spring Boot

Database: MySQL, MongoDB

Web Development: HTML, CSS, JavaScript, Node.js, React

Libraries & Tools: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Streamlit, Git

Cloud: Amazon Web Services (AWS)

Certifications: Data Science and Machine Learning by IIT Guwahati; TensorFlow Keras Bootcamp by OpenCV University

EXTRACURRICULAR

- Achieved 5-star rating in Problem Solving (C++) on HackerRank.
- Actively participated in hackathons such as Smart India Hackathon (SIH), Google Gen AI Hackathon, and Amazon ML Challenge.
- Applied for patent of Predictive Handover of LiFi-WiFi project.